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**Bintanong: An AI Academic Guide for Navigating Curriculum and Institutional Policies at
Palawan State University using Neuro-Symbolic RAG-Based Approach**

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CHAPTER I

INTRODUCTION

As digital technology evolves, properly designed chatbots can provide accessible and context-specific assistance beyond the limited availability of human operators (Villasor et al., 2025). In higher education, academic guidance is important because students need clear information about curricula, course requirements, and institutional policies to make informed decisions and progress toward graduation (Bilquise et al., 2022; Lekan & Pardos, 2025). In the Philippine context, this responsibility is supported by Commission on Higher Education (CHED) Memorandum Order No. 9, s. 2013, which emphasizes effective student welfare and development programs (Siron, 2023).

Students nevertheless continue to rely on Citizens' Charters, printed handbooks, and direct office inquiries. At Caraga State University, 61.3% (n = 245) of students obtained public-service information through frontline staff, while only 17.5% (n = 70) obtained information through the university's Citizen's Charter (Gabia & Urbiztondo, 2025). At Tarlac State University, students identified insufficient discussion of services during orientation and lack of knowledge about service procedures as problems they encountered (Siron, 2023). Students may also hesitate to seek assistance because of embarrassment or fear of judgment (Swacha & Gracel, 2025).

At Palawan State University (PalSU), a local needs assessment involving 202 students found that 55% (111 students) always considered obtaining answers to university-related questions through traditional means difficult and time-consuming (Abela et al., 2025). Although an FAQ website could centralize information, it remains dependent on predefined questions and

answers and may not adequately handle differently worded, ambiguous, multi-step, or conversational inquiries. Previous academic chatbot studies have reported limitations in predefined FAQ-based systems and difficulties with complex or ambiguous queries (Bilquise et al., 2022; Cosicol et al., 2023; Villasor et al., 2025). An AI chatbot provides a distinct advantage by allowing students to ask questions in natural language and receive context-sensitive responses rather than searching for an exact predefined question.

However, generative AI may produce unsupported information when reliable institutional evidence is unavailable (Antico et al., 2024; Iatrellis et al., 2024). Therefore, the advantage of an AI chatbot in this domain depends not only on conversational interaction but also on grounding responses in verified institutional sources and applying explicit rules where institutional decisions require deterministic validation.

To address these limitations, this study proposes *Bintanong*: An AI Academic Guide for Navigating Curriculum and Institutional Policies at Palawan State University using Neuro-Symbolic RAG-Based Approach. *Bintanong* integrates Natural Language Processing (NLP) for English, Filipino, and Taglish queries, a Large Language Model (LLM) for conversational response generation, Retrieval-Augmented Generation (RAG) for retrieving information from authorized PalSU documents, and symbolic reasoning for structured academic and policy rules. *Bintanong* will not autonomously learn missing institutional information from ordinary user conversations; instead, its institutional knowledge will be updated through an authorized document-update process. The study will evaluate whether this integrated pipeline provides more policy-grounded and logically accurate responses than the baseline *Bintu-1* model under controlled evaluation conditions.

Background of the study

Academic guidance influences students' academic performance, retention rates, and overall educational experience, and at PalSU, clear navigation of curriculum and institutional policies is important for student success. When students understand their available academic routes, they are better equipped to make informed decisions and complete their degrees on time. The traditional method of academic information enquiry relies on consulting with course advisors or referencing printed handbooks, such as those containing answers to frequently asked questions (Betrand, Ekwealor, & Onyema, 2023). To address these common advising challenges, researchers are actively deploying Generative AI and Large Language Models (LLMs) to provide real-time, context-aware guidance (Tamascelli et al., 2025) while recent developments in natural language processing have significantly enhanced the capabilities of AI-driven conversational systems (Villasor et al., 2025).

However, current traditional advising practices face systemic challenges, including limited availability, inconsistent guidance, miscommunication, and high administrative workloads for advisors. Academic advisors and student services teams are frequently overwhelmed with queries and cannot provide instant responses, resulting in dissatisfaction with services (Bilquise et al., 2022). Because PalSU operates within this same CHED-compliant administrative framework, it naturally encounters similar systemic challenges. The localized system relies heavily on physical Citizens' Charters posted in front of administrative offices and printed student handbooks—materials that are often difficult for students to fully comprehend. Consequently, staff at the Office of Student Affairs and Services (OSAS) are swamped by repetitive questions, while students are forced to rely on unverified advice from peers, which

slows down office operations and spreads inaccurate information (Abela et al., 2025, Bilquise et al., 2022, Gabia & Urbiztondo, 2025).

But beyond these technological and administrative constraints lies a profound human problem when students face challenges, such as failing a major course, they are often forced to figure out a confusing system on their own by navigating the bureaucratic maze of filing appeals, calculating precise tuition fees, securing an available professor, and enduring the messy encoding procedures of the registrar's system, which had to be done without any official, handed-down guidance. While PalSU's physical Citizens' Charters exist to guide these processes, their effectiveness is limited by high information density and a lack of clear visual hierarchy, making it difficult for students to quickly isolate the answers they need. Their inconsistent layouts and subtle departmental differences render them practically invisible to those in crisis. Yet, students often do not have the capacity to focus on physical books or make the time to meet with their course advisors for guidance. This lack of accessible information inherently affects student performance. Students who need to retake courses cannot properly plan their schedules without enough information, which can lead to inaction until the last minute (Betrand et al., 2023). The emotional toll is severe: the stress of failing is compounded by the fear of appearing uninformed or the anxiety of approaching authority figures (Swacha & Gracel, 2025). This causes many students to remain silent, navigating their academic journey through guesswork.

Furthermore, standard technological solutions have major limitations. Standard LLMs rely entirely on parametric memory, meaning they cannot dynamically update their knowledge without computationally expensive retraining (Lewis et al., 2021; Wu et al., 2024). Because standard LLMs are probabilistic text generators, they are inherently prone to hallucination,

generating highly plausible but factually incorrect statements when their parametric memory is insufficient (Wu et al., 2024). Moreover, academic advising relies on rigid, boolean logic, such as prerequisite dependencies and credit-hour thresholds, which probabilistic LLMs consistently fail to resolve accurately (Lugones et al., 2026). Neural networks fundamentally lack the ability to perform the deterministic, multi-step logical reasoning required for strict rule adherence (Lugones et al., 2026). Additionally, standard LLM tokenizers are highly inefficient for the Philippine context because they do not recognize Tagalog morphological structures, which increases computational cost and degrades semantic understanding (Montalan et al., 2025; Maminta et al., 2025).

Therefore, this study proposes an AI-powered academic guide designed specifically to dismantle these systemic and emotional barriers at PalSU. This proposed solution integrates a multifunctional agent enhanced by a Large Language Model with a Neuro-Symbolic RAG-based approach.

RAG bypasses these parametric limitations by introducing a non-parametric memory component. It utilizes a pre-trained retriever to scan an external, verified database and extract relevant document chunks before passing them to the generator (Wu et al., 2024). The system's RAG pipeline processes institutional documents by chunking and converting them into semantic embeddings, building a searchable knowledge base strictly tied to the uploaded university policies (Nisanth et al., 2025). This allows the external knowledge base to be hot-swapped or updated without needing to retrain the core LLM (Lewis et al., 2021; Wu et al., 2024).

To handle complex academic rules, Neuro-Symbolic AI separates the architecture into two distinct layers: a neural layer for NLP and LLM reasoning, and a symbolic reasoning engine

to process institutional rules and Prolog-based prerequisite validation (Lugones et al., 2026). By routing the decision-making process through the symbolic engine, the system is mathematically constrained to follow programmed institutional rules, significantly reducing the AI's hallucination rate regarding structured policy queries (Lugones et al., 2026). By utilizing Neuro-Symbolic AI, this proposed study guarantees adherence to PalSU's complex institutional rules (Lugones et al., 2026).

Furthermore, localized models optimized for Southeast Asian languages demonstrate markedly improved accuracy in intent recognition and localized context generation (Maminta et al., 2025). Integrating localized Natural Language Processing tuned for Tagalog and English removes the linguistic barriers that standard LLMs exacerbate, ensuring the chatbot communicates in the clear, natural blend of Tagalog and English that students actually use (Maminta et al., 2025; Bilquise et al., 2022).

This proposed study helps to make PalSU's curriculum and institutional policies instantly accessible. By shifting the burden of routine, repetitive procedural inquiries away from human administrative staff, the system reduces the administrative load on the university and the cognitive load on the student (Antico et al., 2024; Kuratomi et al., 2025). Bintanong will serve as a deeply localized, student-centered companion, ensuring that the university's academic processes are efficient, transparent, and that no student must navigate the bureaucracy alone.

Statement of the Problem

Students at Palawan State University (PalSU) may experience difficulty obtaining timely and reliable information about curriculum requirements and institutional policies. Traditional sources such as Citizens' Charters and printed handbooks can be difficult to search and understand, while students may instead rely on repeated office inquiries or unverified peer advice. This problem is supported by Abela et al. (2025), whose local needs assessment found that 55% of surveyed students always considered obtaining university-related answers through traditional means difficult and time-consuming. Previous academic chatbot studies have also reported limitations in predefined FAQ-based systems and difficulties in handling ambiguous or contextually complex queries (Bilquise et al., 2022; Cosicol et al., 2023; Villasor et al., 2025).

Tagalog-English (Taglish), informal, messy, or vague wording may further complicate institutional queries. Studies have reported limitations in Filipino and Tagalog-English language processing and generation (Adoptante et al., 2025; Montalan et al., 2025). RAG also does not automatically guarantee correct answers because response quality depends on the relevance and quality of retrieved context (Antico et al., 2024; Kuratomi et al., 2025).

Institutional advising also involves structured rules such as prerequisite dependencies and eligibility conditions, which should not be determined solely through probabilistic language generation. Iatrellis et al. (2024) reported hallucination risks in generative academic advising, while Lugones et al. (2026) demonstrated the value of symbolic reasoning for enforcing institutional constraints. These limitations provide a basis for evaluating whether the addition of a symbolic reasoning layer to a RAG-based system can improve the handling of institution-specific rules compared with an unaugmented language model.

Therefore, this study seeks to determine whether the full Bintanong pipeline improves policy correctness and reduces unsupported institutional-policy responses compared with the baseline Bintu-1 model under controlled conditions, while also evaluating Taglish handling and the protection of user-uploaded private files.

Specifically, this study seeks to answer the following questions:

1. How can Bintanong be developed by integrating localized Tagalog-English Natural Language Processing, Retrieval-Augmented Generation (RAG), and symbolic reasoning for PalSU institutional queries?
2. Does the full Bintanong pipeline produce more policy-correct and less unsupported responses than the baseline Bintu-1 model when both systems use the same verified query institutional benchmark, prompt template, model version, decoding conditions, and evaluation metrics?
3. How does the Bintanong RAG component perform in terms of retrieval accuracy, context relevance, and response-generation quality when processing queries based on authorized PalSU institutional documents?
4. How accurately does Bintanong handle Tagalog, English, Taglish, messy, and vague institutional queries, including its ability to request clarification or decline unsupported questions?
5. To what extent do Bintanong's private-file safeguards prevent unauthorized access, cross-session exposure, and unintended incorporation of user-uploaded private files into the institutional knowledge base?

6. What is the level of usability and user acceptability of Bintanong among PalSU undergraduate students using standardized evaluation measures?

Objectives of the Study

General Objective

The general objective of this study is to develop and evaluate Bintanong, an AI-driven academic guide integrating localized Tagalog-English Natural Language Processing, Retrieval-Augmented Generation (RAG), and neuro-symbolic reasoning to assist undergraduate students in navigating PalSU curriculum requirements and institutional policies.

Specific Objectives

Specifically, this study aims to:

1. Develop Bintanong by integrating localized Tagalog-English Natural Language Processing, semantic embeddings, K-Nearest Neighbor (KNN) retrieval using Cosine Similarity, Retrieval-Augmented Generation, adaptive query routing, and Prolog-based symbolic reasoning for PalSU institutional queries.

2. Determine whether adding the symbolic reasoning layer to the Bintanong pipeline improves policy correctness and reduces unsupported institutional-policy responses compared with the baseline Bintu-1 model by evaluating both systems using the same verified query institutional benchmark, prompt template, model version, decoding conditions, and evaluation metrics.

3. Measure the RAG component's retrieval accuracy, context relevance, and response-generation quality when processing queries based on authorized PalSU institutional documents.

4. Evaluate Bintanong's ability to accurately interpret and respond to Tagalog, English, Taglish, messy, and vague institutional queries, including appropriate clarification or refusal when sufficient authorized information is unavailable.

5. Evaluate the effectiveness of Bintanong's private-file safeguards in preventing unauthorized access, cross-session exposure, and unintended incorporation of user-uploaded files into the institutional knowledge base.

6. Evaluate the usability and user acceptability of Bintanong among PalSU undergraduate students using standardized evaluation measures.

Significance of the Study

The proposed study aims to advance artificial intelligence in educational contexts and to provide practical insights into the implementation of automated advisory systems using a neuro-symbolic RAG approach.

Palawan State University (PalSU): This proposed study could provide a valuable technological solution for the Office of Student Affairs and Services (OSAS), University and College Student Governments by streamlining the dissemination of academic processes. By implementing this system, the university can reduce the cognitive load for students, decrease the volume of routine inquiries handled by staff, and increase overall student awareness regarding

institutional policies. Therefore, the proposed study contributes to the university's commitment to providing efficient, accessible, and modern student services.

Field of Science. The proposed study contributes to the growing research and application of artificial intelligence in the Philippines. Specifically, it sheds light on the practical implementation of the neuro-symbolic RAG approach within an undergraduate setting. By applying this advanced AI architecture to a university context, the proposed study explores new frontiers in educational technology and opens up broader opportunities for growth and innovation in the local AI landscape.

Educational Value. The proposed study presents a comprehensive example of applying advanced artificial intelligence models to practical, real-world institutional problems. It serves as an invaluable educational resource for Filipino students and AI researchers by demonstrating a scalable concept. The theoretical and practical frameworks presented in this proposed study illustrate how AI can solve complex information-retrieval problems, a concept that can be adapted and applied to a variety of vital sectors, including disaster risk reduction, legal frameworks, and digital governance.

Future Researchers. The application of chatbot guidance to both institutional policies and navigating curriculum in this proposed study contributes to the current understanding of AI utility in academic environments. This work can serve as a foundational basis for future AI researchers in the Philippines, encouraging further studies, refinements, and innovations in automated advisory systems and neuro-symbolic RAG applications.

Scope

This proposed research study focuses on the development and evaluation of Bintanong, an AI-powered academic guide for undergraduate students at PalSU during the 2026–2027 academic period. The system will center entirely on navigating curriculum and institutional policies, translating complex rules from the curriculum, PalSU Student Handbook, Citizens' Charter, USG and CSG Policies into actionable steps for student transactions. Furthermore, the system will be tailored exclusively for undergraduate academic affairs.

Crucially, Bintanong will be bound as a read-only advisory tool. To ensure policy accuracy and prevent data degradation, the system will not autonomously learn from ongoing user interactions or automatically acquire information that is absent from its knowledge base. Instead, updates to its institutional knowledge will only occur through the addition and verification of authorized institutional documents by the researchers or authorized personnel.

Additionally, Bintanong will be developed exclusively as a web-based platform, prioritizing cross-platform accessibility via modern web browsers. To safely process its knowledge base, the system utilizes a hybrid Neuro-Symbolic RAG architecture. This pairs a neural component and a localized Tagalog and English NLP model to handle unstructured conversations and Taglish user queries, alongside a symbolic engine to strictly follow structured university logic.

The system will specifically accommodate institutional queries expressed in English, Filipino, and Taglish. Questions that are unrelated to the authorized institutional sources or cannot be sufficiently addressed using the supported source language and available institutional information will be treated as out of scope. In such cases, Bintanong will be expected to request

clarification, indicate that the information is unavailable, or refer the user to the appropriate university office rather than generate an unsupported institutional response.

For user-uploaded private files, Bintanong will implement preventive measures to keep these files separate from the institutional knowledge base. Authorized private files will be treated as user-specific information and will not be automatically indexed or incorporated into the institutional knowledge base. Access will be restricted to the authorized user/session, and the system will be evaluated for unauthorized access, cross-session exposure, and unintended incorporation of private-file contents into institutional retrieval.

Limitations

Despite its proposed contributions, the implementation of the system is subject to several operational boundaries. While the undergraduate knowledge base is covered in this proposed study, it inherently excludes procedures for faculty, staff, laboratory school, and professional degree programs such as Medicine and Law. Additionally, AI will rely entirely on the digitized documents provided; if a specific policy, memorandum, or other institutional information is missing from the system's authorized knowledge base, Bintanong will not automatically learn or acquire the missing information from user interactions. Consequently, the system may be unable to provide an institution-specific answer and will instead be expected to indicate that the information is unavailable, request clarification, or refer the user to the appropriate university office. Moreover, Bintanong lacks the security clearance or write access required to alter the university's central database, so it cannot perform administrative tasks such as officially enrolling a student, printing a Certificate of Registration (COR), or updating grades.

The protection of private user-uploaded files is limited to the security controls

implemented and tested within the Bintanong system. These controls include separating private-file contents from the institutional knowledge base, restricting access to the authorized user/session, preventing automatic indexing of private files into the institutional retrieval database, and testing whether another user or session can retrieve the contents of a private file. The study does not claim protection against security threats beyond the system components and test conditions covered by the research. Technologically, users may experience processing latency due to the sequential execution of the neural and symbolic components, particularly the inference overhead required for intent extraction and final response generation. This delay is further extended when a policy-bound query activates both the RAG retrieval pipeline and the Prolog symbolic engine, as the system must wait for both to finish before a response can be generated. If the output validation fails, the language model regenerates the response from the beginning, adding more waiting time on top of the initial delay. For online deployment of Bintanong to a cloud server, the system does not account for traffic/network disruptions coming from the user side (i.e., weak Internet signal), and does not account for cloud server disruptions. Furthermore, concurrent user requests may result in significant processing bottlenecks.

Due to financial constraints typical of student-led research, the system will be developed with a typical computational hardware stack found in entry-level AI student-led research. Hence, the NVIDIA GeForce RTX 4060 Laptop GPU will be considered for this research, along with the model choice for Bintu-1, to simulate an LLM with enough parameters to fit within the GPU VRAM and avoid falling back to system RAM. To further constrain Bintanong within the researchers' financial capability, the system will be deployed on entry-tier cloud hosting services, which possess limited computational bandwidth and rate limits to accommodate high-volume, simultaneous traffic. Finally, although Bintanong is designed to process Tagalog, English, and

Taglish queries, its ability to provide an accurate institutional response remains limited when a query contains insufficient information, is outside the supported institutional scope, or requires information that is not present in the authorized knowledge base. Rather than assuming missing information, the system is designed to request clarification or decline to provide an unsupported answer.

Operational Definition of Terms

Adaptive Query Routing: Refers to the mechanism within Bintanong's architecture that classifies an incoming student's prompt by its complexity. It routes open-ended conversational questions to the neural language model and directs strict, policy-bound questions to the symbolic reasoning engine to ensure computational efficiency and factual accuracy.

Bintu-1: Refers to the local name of the baseline LLM that the Bintanong will utilize. This specifically identifies the [Google Gemma4 e4b](#) quantized model, which serves as the core generative engine of the system.

Chunks: Refers to the smaller, segmented passages that institutional documents such as the PalSU Student Handbook, Citizens' Charter, USG and CSG Policies are broken down into during preprocessing, allowing the retrieval system to extract and return only the most relevant portions of a document in response to a student query.

Closed World Assumption (CWA): Refers to the presumption within the system's knowledge representation that the encoded PalSU policies and curriculum needed for reasoning are present within the current PalSU scope.

Code-Switching (Taglish): Refers to the authentic communicative pattern of alternating between Tagalog and English. In this proposed study, it represents the specific linguistic challenge that standard AI models fail to process, necessitating Bintanong's localized natural language understanding capabilities.

Cognitive Load: Refers to the measurable mental effort and working memory demands placed on a student when interacting with the chatbot or navigating PalSU processes and policies, which this proposed study aims to evaluate and minimize.

Corpus: Refers to a static, predefined dataset of text or conversational pairs used to train earlier generations of AI chatbots. This proposed study intentionally moves away from reliance on a static corpus to avoid knowledge staleness when PalSU policies are updated.

Cosine Similarity: Refers to the metric used to measure the contextual alignment between a student's query vector and the document vectors in the knowledge base by computing the cosine of the angle between two non-zero vectors.

Digital Modality: Refers to the electronic platform (e.g., a web-based or mobile application) through which the Bintanong chatbot is deployed and accessed by the students, as opposed to traditional face-to-face transactions.

Fine-Tuning: Refers to the method of adapting a general-purpose AI by updating its internal neural weights using domain-specific data. This proposed study eschews fine-tuning in favor of RAG to avoid the risks of overfitting, repetitive generation, and the computational cost of continuous retraining when institutional rules change.

Hard-Constraint Logic: Refers to the strict, uncompromising rules governing PalSU's academic policies, such as prerequisite dependencies, graduation eligibility, and credit unit thresholds. Bintanong uses a symbolic reasoning layer to enforce these hard constraints, as neural models cannot reliably compute them.

Hot-swapped (or Hot-swap): Refers to the system's technical capability to update or replace its underlying knowledge base (such as institutional policies) or AI models without shutting down the chatbot or interrupting student access.

K-Nearest Neighbor (KNN): Refers to the machine learning algorithm used in Bintanong's retrieval pipeline to identify and extract the top-k most contextually relevant document chunks from the vector database using Cosine Similarity as its distance metric.

Natural Language Processing (NLP): Refers to the computational techniques implemented in the chatbot to parse, interpret, and extract intent from students' text-based inquiries.

Neuro-symbolic AI: Refers to the hybrid architecture employed in this proposed study that combines the LLM's neural network with the structured, rule-based logic of PalSU academic and clients policies, ensuring the chatbot provides factually grounded answers with minimal risk of hallucination.

Non-Parametric Memory: Refers to the external, dynamically updatable knowledge base of the system, specifically the indexed vector database containing the PalSU Student Handbook, Citizens' Charter, USG and CSG Policies —from which Bintanong retrieves authoritative facts.

Optical Character Recognition: Refers to the ability of a computer to identify the optical characters and converts the text content from a picture to a computer readable text. In Bintanong's architecture, this component is used to parse the text information from photographed Citizen's Charters and other PalSU policies into retrieval ready text.

Parametric Memory: Refers to the knowledge stored implicitly within the internal neural weights of the Large Language Model. In Bintanong's architecture, this component is restricted to handling fluid language interpretation and generation, rather than acting as the source of factual institutional truth.

Pipeline: Refers to the automated, step-by-step technical workflow of the system, starting from the moment a student inputs a prompt, through tokenization and retrieval (RAG), to the generation of the final response.

Predicate: Refers to the formal representation of a property or relation applied to an object within a First-Order Logic expression, used in Bintanong to encode institutional conditions into mathematically verifiable logical statements.

Prolog: Refers to the declarative logic programming language used to implement Bintanong's symbolic reasoning engine, integrated into the Python-based architecture through frameworks such as PySwip or Janus.

Reinforcement Learning from Human Feedback (RLHF): Refers to a training technique that aligns Large Language Models with human preferences by incorporating human evaluations as a reward signal, influencing how the model responds to queries, including its tendency to refuse answering rather than risk generating hallucinated information.

Retrieval-Augmented Generation (RAG): Refers to the specific framework integrated into the chatbot that forces the AI to search and extract facts from the database of PalSU documents before answering a student's query.

Self-Attention: Refers to the core mechanism of the Transformer architecture that allows the LLM to dynamically weigh the relevance of every part of the input, including the student's query and retrieved policy passages, when generating each token of the response.

Semantic Embeddings: Refers to the mathematical vector representations of PalSU's documents and student queries, allowing the RAG system to match a student's question to the correct policy based on contextual meaning rather than exact keyword matches.

Symbolic Reasoning: Refers to the explicit, rule-based logic processing component of Bintanong's hybrid architecture. Unlike opaque neural networks, this component enforces structural and procedural correctness, providing an auditable and verifiable trail for how an academic guidance answer was formulated.

Tokenizer: The algorithmic tool used in the chatbot's preprocessing phase to break down PalSU students' queries into smaller, machine-readable pieces for more efficient AI processing.

Top-k Retrieval: Refers to the selection phase of the KNN algorithm where exactly k document chunks with the highest similarity scores against the student's query vector are extracted and passed into the LLM's generation process.

Transformer Architecture: Refers to the deep learning architecture underlying the Large Language Model used in Bintanong, such as Gemma4 e4b, which processes input through

a self-attention mechanism to capture contextual relationships between the student's query and the retrieved policy passages before generating a response.

Universal Quantifier ($\forall$): Refers to the logical operator in First-Order Logic expressing that a rule applies to all members of a defined group, used in Bintanong to encode policies that apply universally to all students meeting a specific condition.

Vector Embeddings: Refers to the numerical vector representations generated from text, both from the institutional documents and the student's query, that capture the semantic meaning of the content and allow the system to compare and retrieve contextually relevant information within the vector space.

Vector Space: Refers to the high-dimensional mathematical space in which both the student's query and the institutional document chunks are represented as vectors, enabling the KNN algorithm to measure their semantic similarity through Cosine Similarity computations.

Vector Store: Refers to the component that stores, indexes, and queries the high-dimensional semantic vector representations of institutional documents such as the PalSU Student Handbook, Citizens' Charter, USG and CSG Policies enabling the KNN engine to efficiently execute similarity based retrieval during runtime. In Bintanong, this is to be implemented using tools, specifically pgvector.

Zero-Shot Decoding: Refers to a scenario where an AI model is forced to answer a query using only its pre-trained knowledge, without being provided any external context, documents, or examples. This proposed study utilizes RAG specifically to prevent Bintanong from operating in a zero shot environment when answering critical PalSU policy questions.

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CHAPTER III

THEORETICAL BACKGROUND

This chapter presents the background and characteristics of the computational concepts related to the proposed study. It explores the functionality and purpose of each of the computational concepts' characteristics and expands it to understand the role of each.

Sentence Embedding Model

When humans speak about something, like a statement that connects a singular object into a single phrase in this example: **“I had two apples today, a red, juicy, sweet one, and also the latest orange iPhone 17 Pro Max 1TB one.”**, humans will use **context clues** in order to find the similarity of the following premises:

1. The red, juicy, sweet one is an actual fruit called **“apple”**
2. The other orange one is the color of the latest iPhone released by **“Apple”** Inc.

This single sentence has the context similarity of a single word “apple”, since one is an actual fruit, while the other one, Apple Inc., is a major American technology company that produces electronics like the **iPhone 17 Pro Max**.

While humans use context clues in natural language in processing and understanding the semantic meaning of the word “apple”, in computing, words **cannot be directly processed** as **natural language** without some of conversion from keywords to a numerical representation, as numbers are the only representation the core LLM of Bantanong (Bintu-1) is familiar with. This is where the concept of **sentence embedding** enters the picture in pre-processing queries.

Sentence embedding is a Natural Language Processing technique that converts a complete sentence or text passage into a single fixed-size numerical vector that captures its semantic meaning. Unlike traditional keyword-based representations that treat words as isolated units, sentence embeddings encode the contextual relationships between words so that sentences with similar meanings produce vectors that are close to each other in the vector space, regardless of whether they use the exact same words.

The general process of producing a sentence embedding follows a sequential pipeline of five operations.

Step 1: Tokenization

Given a raw input sentence S , the SentencePiece tokenizer breaks it down into a sequence of n subword units called tokens:

$$T = [t_1, t_2, t_3, \dots, t_n]$$

SentencePiece is specifically utilized because it is the native tokenization algorithm of the SEA-LION-Embedding-E5-600M model, enabling it to efficiently handle rare words, misspellings, and morphologically complex Taglish words by breaking them into smaller, recognizable pieces.

Step 2: Token Embedding

Each token t_i is mapped to a dense numerical vector e_i of dimension d through a learned embedding matrix E :

$$e_i = E(t_i) \in \mathbb{R}^d$$

This produces a sequence of token vectors:

$$[e_1, e_2, e_3, \dots, e_{\square}]$$

At this stage, each token vector only captures the meaning of the individual token and does not yet reflect the surrounding context.

Step 3: Contextual Encoding via Transformer Layers

The sequence of token embeddings is passed through multiple Transformer layers. Inside each layer, the Self-Attention mechanism allows every token to attend to every other token in the sequence, updating each token's representation to reflect its meaning given its full surrounding context. After passing through all layers, each token holds a contextually enriched hidden state h_i :

$$h_1, h_2, \dots, h_{\square} = \text{Transformer}([e_1, e_2, \dots, e_{\square}])$$

Step 4: Mean Pooling

Since the Transformer produces one vector per token, the model aggregates the entire sequence into a single sentence-level vector through mean pooling, which computes the average of all token hidden states:

$$\vec{s} = \frac{1}{n} \sum_{i=1}^n \vec{h}_i$$

Where $\vec{s} \in \mathbb{R}^d$ is the pooled sentence representation.

Step 5: L2 Normalization

The pooled vector is normalized to unit length so that all sentence vectors lie on the surface of a unit hypersphere. This ensures that similarity comparisons using Cosine Similarity are based purely on the direction of the vectors rather than their magnitude:

$$\hat{s} = \frac{\vec{s}}{\|\vec{s}\|_2}$$

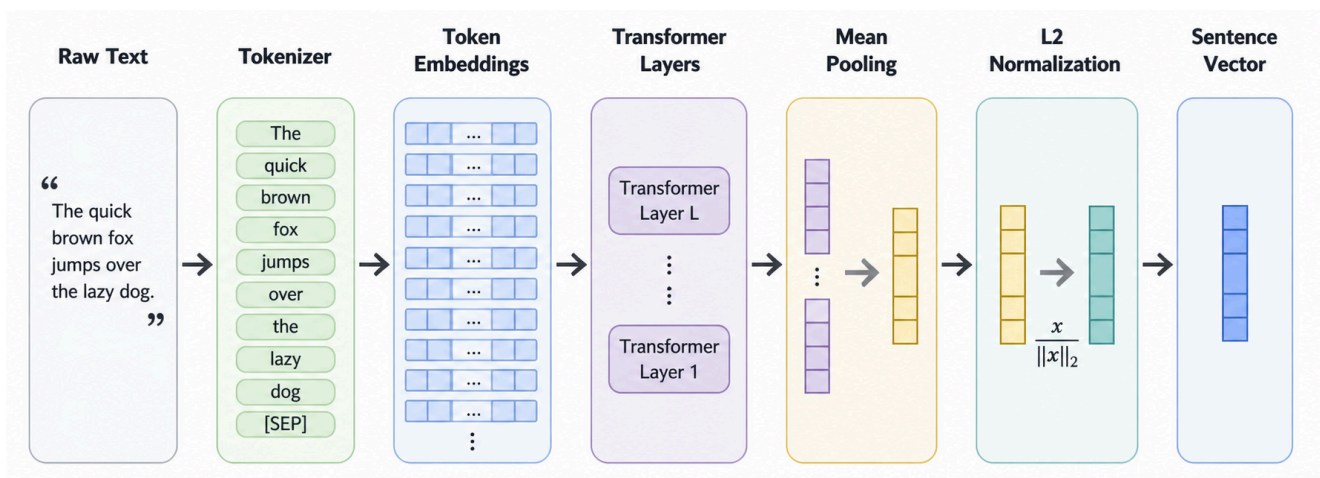


Figure 3.1: General Sentence Embedding Pipeline

Alignment with the Core Research Problem

The PalSU Student Handbook, Citizens' Charter, USG and CSG Policies are written in formal English, while PalSU students naturally communicate in Taglish. Traditional keyword matching falls short in this context because a student's naturally phrased Taglish query and the formal English policy text rarely share the same exact vocabulary, even when they refer to the same topic. The sentence embedding model, SEA-LION-Embedding-E5-600M, addresses this directly by encoding meaning rather than keywords, so that a student asking "Pwede ko bang kunin ang CC5 kahit bumagsak ako sa CC4?" produces a query vector that lands semantically close to the English policy passage describing prerequisite requirements. This cross-lingual semantic matching is the specific technical contribution of Bintanong's NLP layer, removing the linguistic barrier that standard monolingual systems would impose on Filipino students navigating institutional policies.

Cosine Similarity

Metric that measures the similarity between two non-zero vectors by calculating the cosine of the angle between them. It is widely used in machine learning and data analysis, especially in text analysis, document comparison, search queries, and recommendation systems.

$$\cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|} = \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2} \sqrt{\sum_{i=1}^n B_i^2}}$$

The Cosine Similarity formula, expressed as $\cos(\theta)$, mathematically measures the relevance between a student query (**A**) and a specific handbook policy (**B**). The process will begin by analyzing the total dimensions (***n***), which represent the various topics or concepts within the system, using the current data index (***i***) to navigate through each one. At each index, the system will compare the individual weights (***A_i*** and ***B_i***) to determine how closely the student's intent matches the handbook's content for that topic. By summing these values (Σ), the system will identify the shared keyword points (**$A \cdot B$**), also known as the dot product. Finally, this total is divided by the text-length normalization ($\|A\| \|B\|$) to ensure that the final similarity score remains accurate and fair, regardless of whether the student's question is a short phrase or a long sentence.

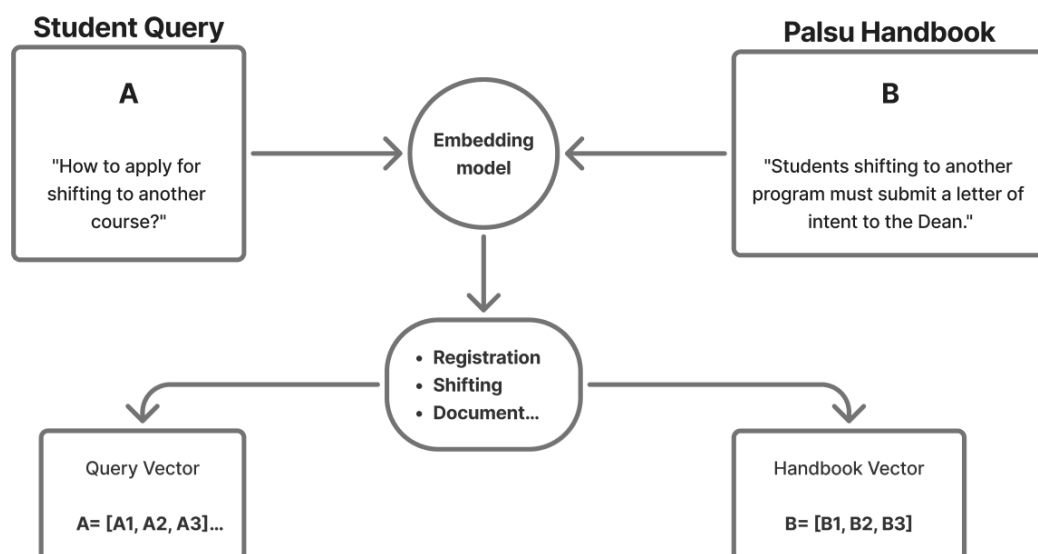


Figure 3.2: Cosine Similarity Simulation

Calculation:

$$\cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|} = \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2} \sqrt{\sum_{i=1}^n B_i^2}}$$

$\mathbf{A} \cdot \mathbf{B}$:

$$(\mathbf{A1} \cdot \mathbf{B1}) + (\mathbf{A2} \cdot \mathbf{B2}) + (\mathbf{A3} \cdot \mathbf{B3}) = \mathbf{AB}$$

$(\|\mathbf{A}\| \|\mathbf{B}\|)$:

$$\mathbf{A} : \sqrt{\mathbf{A1}^2 + \mathbf{A2}^2 + \mathbf{A3}^2} = \sqrt{\mathbf{A4} + \mathbf{A5} + \mathbf{A6}} = \sqrt{\mathbf{A7}} \approx \mathbf{Ai}$$

$$\mathbf{B} : \sqrt{\mathbf{B1}^2 + \mathbf{B2}^2 + \mathbf{B3}^2} = \sqrt{\mathbf{B4} + \mathbf{B5} + \mathbf{B6}} = \sqrt{\mathbf{B7}} \approx \mathbf{Bi}$$

$$\cos(\theta) = \frac{\mathbf{AB}}{\mathbf{Ai} \cdot \mathbf{Bi}}$$

If $\cos(\theta) = 1$, the angle is 0° , meaning the student query and the handbook policy are a perfect match. If $\cos(\theta) = 0$, the angle is 90° , meaning the two topics are totally unrelated and share no common meaning. If $\cos(\theta) = -1$, the angle is 180° , meaning the concepts are complete opposites in the data space.

K-Nearest Neighbor (KNN) Algorithm

K-Nearest Neighbor (KNN) is a simple and widely used machine learning technique for classification and regression tasks. It works by identifying the K closest data points to a given input and making predictions based on the majority class or average value of those neighbors.

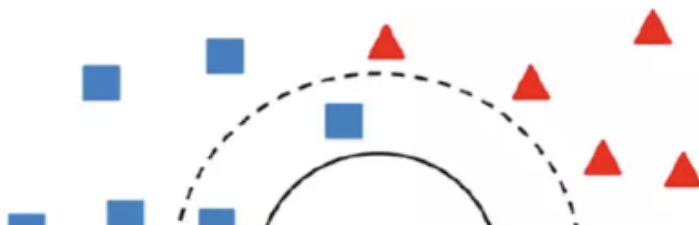


Figure 3.3: Traditional K-Nearest Neighbor

To determine which data points are the "closest," traditional KNN relies on spatial metrics, the most common being **Euclidean Distance**. This algorithm measures the direct, straight-line physical distance between two points in a multidimensional space. It is mathematically expressed as:

$$d(\mathbf{x}, \mathbf{X}_i) = \sqrt{\sum_{j=1}^n (x_j - X_{ij})^2}$$

Where $d(\mathbf{x}, \mathbf{X}_i)$ is the calculated spatial distance between the unclassified target input ($\mathbf{x}$) and a specific historical data point residing in the existing dataset ($\mathbf{X}_i$). To determine this spatial proximity, the algorithm evaluates the data across n , which defines the total number of dimensions or features being measured within the data space. During this calculation, the variables x_j and X_{ij} act as the specific numerical values of the j -th feature for both the target input

and the stored data point, respectively, allowing the system to quantify the precise mathematical difference at every individual dimension before aggregating the final distance.

K-Nearest Neighbors (KNN) via Cosine Similarity (Retrieval)

While traditional KNN relies on spatial distance measurements, applying KNN for semantic search and Retrieval-Augmented Generation (RAG) requires identifying the most relevant information by evaluating the target query against all available documents in the vector space based on contextual alignment. Instead of calculating Euclidean spatial distance, the algorithm computes the Cosine Similarity for every query-document pair, ranks them, and extracts the top- k semantic vectors that share the highest similarity orientation with the query.

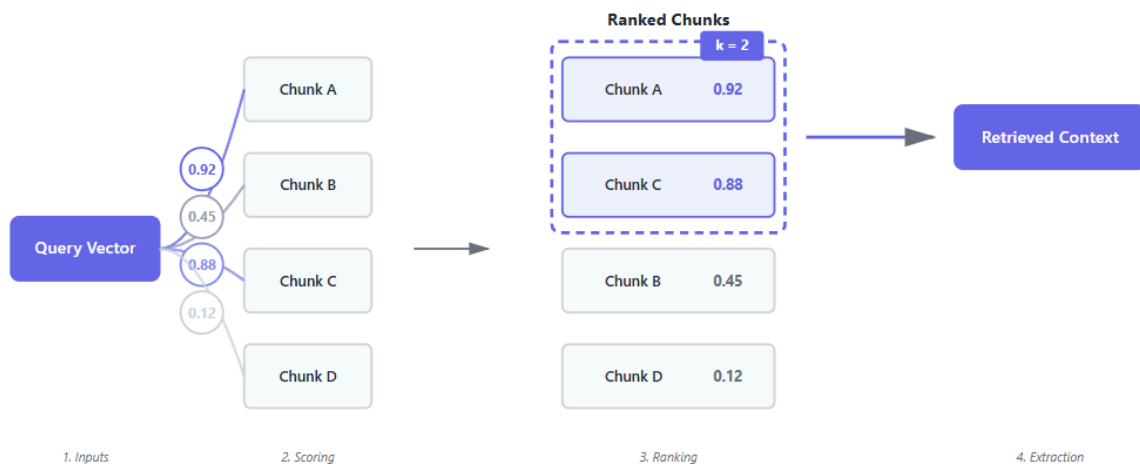


Figure 3.4: Top-K Retrieval and Ranking

The final phase of the algorithm, as shown in Figure 3.4, involves ranking the evaluated query-document pairs and extracting the most relevant data. Once all similarity scores have been

computed between the query and the database chunks, the algorithm isolates the highest-ranking vectors using the following expression:

$$\text{Top-}k \text{ Neighbors} = \underset{\mathbf{d} \in D}{\operatorname{argmax}}^k (\text{Similarity}(\mathbf{q}, \mathbf{d}))$$

Where k is the predefined number of nearest neighbors (most relevant chunks) the algorithm will retrieve to form the final context. The argmax^k function acts as a sorting and selection mechanism, searching through all evaluated documents in the database (D) to extract exactly k chunks that yielded the maximum similarity scores when compared against the semantic vector of the student's query (q).

Implementation in Code and Data Processing

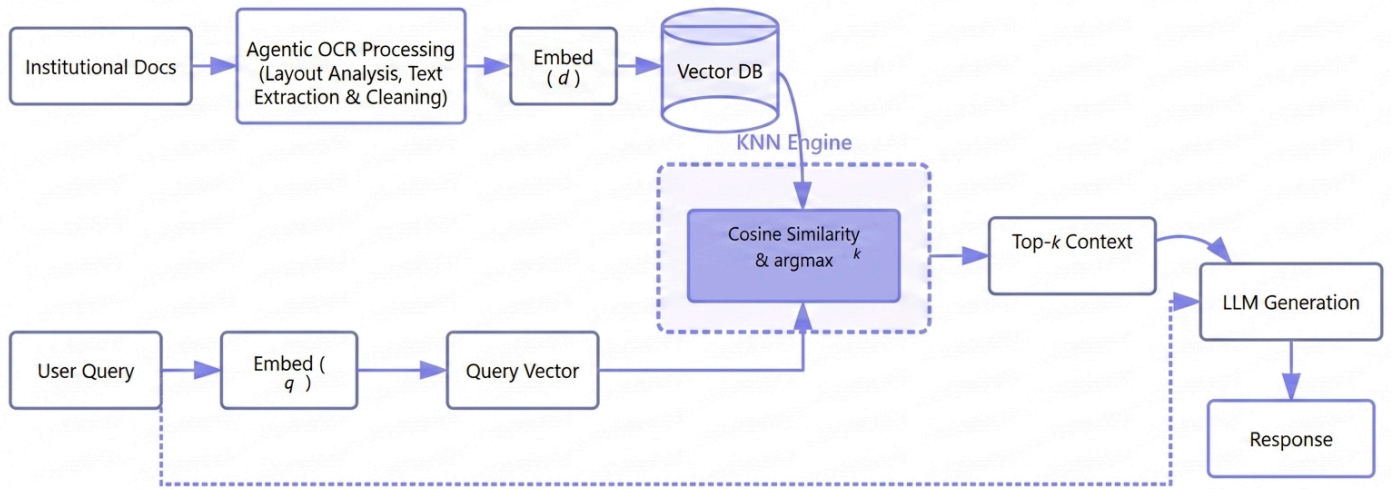


Figure 3.5: Technical Workflow of the KNN-based Retrieval and Data Processing Pipeline

The theoretical mechanism of K-Nearest Neighbors via Cosine Similarity will be directly translated into Bintanong's web-based data retrieval pipeline, as illustrated in Figure 3.5. Textual

data derived strictly from the PalSU Student Handbook, Citizens' Charter, USG and CSG Policies will be processed through an agentic OCR system for layout analysis, text extraction, and cleaning, then pre-processed, chunked, and converted into high-dimensional semantic vectors. These document vectors (d) will be persistently stored within the system's vector database.

During runtime, when a student submits an inquiry (Taglish or English) via the web interface, the backend system will embed the query into the same vector space (q). The vector store will then execute the KNN algorithm natively, calculating the cosine similarity between the query vector and the stored institutional documents. By programmatically executing the argmax_k function, the system will filter and extract the top-k most relevant text chunks. These retrieved documents will subsequently be routed into the context window of Bintu-1 to ground its generation.

Alignment with the Core Research Problem

This retrieval algorithm will directly address the core objective of developing an accurate, institution-specific academic guide by grounding Bintanong strictly in verified institutional policies. By incorporating Retrieval-Augmented Generation (RAG) and leveraging KNN context retrieval accuracy, relevance, and response quality (Objective 3), the system will ensure that whenever a student asks about PalSU curriculum or policies, the AI will not rely on generalized pre-trained weights or invent facts. Instead, the algorithm will enforce that the system retrieves and restricts its answers to the exact documentation pulled from the PalSU

Student Handbook, Citizens' Charter, USG and CSG Policies. This mechanism will mathematically guarantee that the academic guidance provided will be reliable and consistent with the university's official frameworks, thereby enabling logical accuracy and reducing hallucinations when benchmarked against the baseline version of Bintu-1.

First-Order Logic (Predicate Logic)

The First-Order Logic, or more simply called Predicate Logic, is a formal system of representing complex knowledge and object reasoning in the fields of mathematics, philosophy, and computer science, where it is an extension of its base foundation, propositional logic (Manduzio [et.al](#), 2024). First-Order Logic (FOL) extends this by dividing statements into subjects or objects and predicates, more informally known as relations or properties relevant to the objects, with quantifiers like "all" or "some" setting the scope of those relations, enabling reasoning about groups of objects rather than just individual statements.

First-Order Logic relates to how humans reason because people often understand the world by identifying objects, assigning meaning to them, recognizing their properties, and connecting them to other objects through relationships (Stol, M.C., & Mileo, A, 2024).

In everyday reasoning, the human mind does not treat knowledge as isolated statements only; instead, it organizes information based on entities, categories, conditions, and rules. For example, a person may reason that a student can enroll in a subject only if that student has passed the required prerequisite. This kind of thinking follows a structured pattern where objects are identified, their conditions are checked, and conclusions are drawn from established rules. In this sense, First-Order Logic does not fully replicate the human mind, but it provides a formal way to represent the rule-based and relational aspects of human reasoning.

For example, consider the following statement:

“Sandrone is a math wizard; therefore, she is the best to teach Computational Sciences.”

In its raw form, this introduces a singular object ("Sandrone") and a predicate ("MathWizard"), but it lacks a universal rule to make the inference logically complete. To properly form the argument, a general rule must be added:

"All math wizards are the best to teach Computational Sciences."

In this case, now the entire structure of the argument, English-formal format that is, now looks like this:

- ***“Sandrone is a math wizard; all math wizards are best to teach Computational Sciences, therefore, she is the best to teach Computational Sciences.”***

In FOL, it can be represented as:

- “For every x as a person, if x is a math wizard, then x is best to teach CompSci.”

$$\forall x(\text{MathWizard}(x) \rightarrow \text{BestToTeachCompSci}(x))$$

In this FOL format, as shown in expression above, it can be derived as:

- $\forall x$ = "for every" person x
- MathWizard(x) = the predicate applied to x
- $\rightarrow$ as then or implication
- BestToTeachCompSci as the conclusion

So the statement:

“Sandrone is a math wizard; all math wizards are best to teach Computational Sciences, therefore, she is the best to teach Computational Sciences.”

Is considered:

VALID

Why valid, someone might ask?

This argument is considered valid in FOL, meaning the structure is logically sound given the stated premises. In this formal system, the same structural template directly maps to how PalSU's institutional rules are encoded in Bintanong: a universal rule, a set of known facts about a specific student, and a deterministic conclusion derived from them.

Research Implementation:

First-Order Logic will be implemented in Bintanong through Prolog, a declarative logic programming language used for encoding facts, rules, and inference-based reasoning. Since Bintanong's main architecture is Python-based, Prolog will be integrated through a Python-to-Prolog middleware layer such as Janus or PySwip. This allows the adaptive query router to send structured academic queries to the symbolic reasoning layer whenever logical validation is required.

During operation, the symbolic layer will be activated in two major cases. First, it will handle queries that can be directly resolved through encoded academic rules, such as prerequisite checking, subject eligibility, and course dependency validation. In these cases, the system can

query the Prolog knowledge base directly instead of relying on full document retrieval, reducing unnecessary computational workload.

Second, the symbolic layer will act as a guardrail when an open-ended student query also involves structured academic rules. For example, if a student asks how to recover after failing a foundational subject such as CC4/L, the neural and retrieval layers may help explain the situation conversationally, but the symbolic layer verifies which subjects are affected by the failed prerequisite. This ensures that the LLM’s final response remains constrained by institutionally valid rules rather than by statistically plausible but potentially incorrect generation.

Alignment with the Core Research Problem:

First-Order Logic functions as Bintanong’s symbolic reasoning layer, providing a deterministic and mathematically grounded way to validate structured academic rules. While RAG helps retrieve relevant institutional information and the LLM translates that information into natural-language responses, these components alone cannot fully guarantee correctness because LLMs remain probabilistic and may still hallucinate, overgeneralize, or respond with uncertainty. By encoding policy-bound rules such as prerequisites, subject retaking conditions, and academic procedures through FOL and Prolog-based inference, Bintanong ensures that answers involving institutional rules are logically verified before being presented to students. This allows the system to combine the conversational flexibility of LLMs with the consistency, auditability, and reliability required in high-stakes academic guidance.

As a preliminary illustration of how the LLM will function with a symbolic, first-order logic toolchain, let's consider a student who asks Bintanong an academic-bound question: *"Puwede ko pa bang kuhanin ang CC5 kahit bumagsak ako sa CC4?" (Can I still take CC5 even if I failed CC4?)*

The adaptive query router classifies this as a structured, policy-bound query and routes it directly to the symbolic layer. In an example implementation inside SWI-Prolog, the institutional rule is encoded as:

```
prerequisite(cc5l, cc3l).
```

```
prerequisite(cc5l, cc4l).
```

```
failed(student_x, cc4l).
```

```
passed(student_x, cc3l).
```

```
% Rule: student can enroll if ALL prerequisites are passed
```

```
can_enroll(Student, Subject) :-
```

```
\+ (
```

```
    prerequisite(Subject, Prereq),
```

```
    \+ passed(Student, Prereq)
```

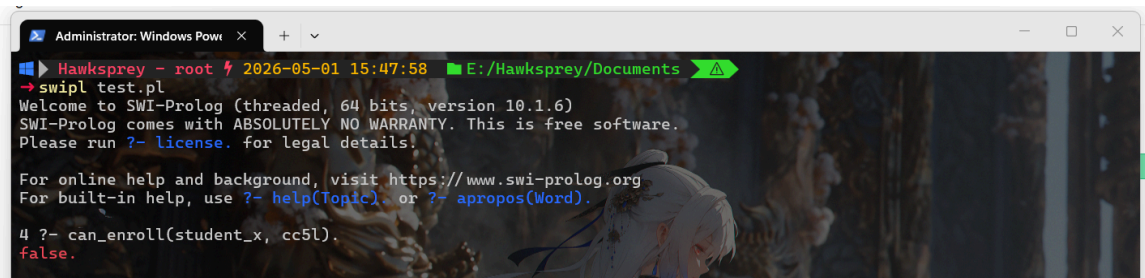
```
).
```

In this sample knowledge base, CC5/L, or Information Management, requires both CC3/L and CC4/L as prerequisites. The student is encoded as having passed CC3/L but failed CC4/L. The `can_enroll(Student, Subject)` rule then checks whether all required prerequisites for

the subject have a matching **passed** record. If even one prerequisite is missing or not proven as passed, the rule returns false.

Applied to the sample query, Prolog cannot prove that **student_x** passed CC4/L. Because of this, **can_enroll(student_x, cc5l)** evaluates as false, meaning the student is not eligible to take CC5/L. This follows the same basic structure as conjunctions in propositional logic: the student must pass both CC3/L and CC4/L. If either condition is false, the overall eligibility condition is also false.

Then it would be considered as **false** (or “Therefore, the student is not eligible to take CC5/L”):



```
Administrator: Windows PowerShell
Hawksprey - root 2026-05-01 15:47:58 E:/Hawksprey/Documents
-> swipl test.pl
Welcome to SWI-Prolog (threaded, 64 bits, version 10.1.6)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details!

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

4 ?- can_enroll(student_x, cc5l).
false.
```

*Figure 3.6: Prolog sample code and output, when the parameters of **student_x** and **cc5l** are run with the encoded knowledge in SWI-Prolog in Windows Terminal.*

This is where **Negation as Failure** in Prolog enters the process. In simple terms, if Prolog cannot prove that a required condition is true within its knowledge base, it treats that condition as false. In the context of the sample query, this can be understood as: **“If the system cannot prove that the student has passed CC4/L or CC3/L, then the student cannot be considered eligible to take CC5/L.”**

In Bintanong’s context, the `can_enroll(Student, Subject)` rule checks the encoded facts in the knowledge base, specifically the `prerequisite` and `passed` records. For the rule to return true, all prerequisites must have a matching passed record. If one prerequisite is missing, failed, or not proven as passed, the rule returns false. This is similar to conjunction in propositional logic, shown below, where all required conditions must be true for the whole statement to be true.

Example:

“A student needs to both pass CC4/L (p) and CC3/L (q) in order to take CC5/L or Information Management.”

p: Passed CC4/L (Data Structures and Algorithms)

q: Passed CC3/L (Computer Programming 2)

Presented as $P \wedge Q$:

<i>A student needs to both pass CC4/L (p) and CC3/L (q) in order to take CC5/L or Information Management.</i>	Passed CC4/L (Data Structures and Algorithms) as (P)	Passed CC3/L (Computer Programming 2) as (Q)	Answer for the proposition of $P \wedge Q$	Results
	True	True	True	You can take CC5/L
	True	False	False	You can’t take CC5/L
	False	True	False	You can’t take CC5/L
	False	False	False	You can’t take CC5/L

Table 3.1: A truth table constructed for the propositional logic validity of CC5/L’s prerequisite chain

In the sample Prolog query, the system cannot find a valid fact for `passed(student_x, cc4l)`. Because of the `\+` operator, Prolog treats the unproven condition as unmet. Under the **Closed World Assumption**, any fact that cannot be found or proven inside the current knowledge base is treated as false. Therefore, `can_enroll(student_x, cc5l)` returns false because the student has not satisfied all prerequisite conditions.

In Bantanong’s architecture, the symbolic architecture layer is a deliberate choice for presenting first-order logic within the context of an LLM, since instead of generating a response that is plausible, but logically invalid due to the LLM’s probabilistic nature, it deflects it by grounding the response in a logical layer, and in the context of the sample query, the system defaults to deny or deflecting the query rather than allow when institutional records are incomplete, ensuring that no student could accidentally think they are eligible for a subject they have not legitimately completed. This makes the symbolic layer a mathematically hard constraint on the system, one that cannot be overridden by the probabilistic tendencies of the underlying language model, no matter how confidently the LLM might otherwise generate an incorrect advisory response.

Large Language Model

To understand how a Large Language Model parses and understands information, it is essential to break down how human brains process information first. In human brains, any form of input, especially in the context of natural language, starts with the following sentence example: **“I found out that since Martin and John represented the university as Tribu**

Haliya in the last eSports tournament, did they actually crush the defending team during the finals?”

To parse that information in natural language, a human brain would do the following:

- Finding the keywords and context clues by recognizing the keywords and patterns involved and associating them (example: eSports -> any competitive online game; university- > (Tribu Haliya)
- Retrieve the highlighted patterns and keywords from the human brain if the knowledge does exist in one form or another
 - If the human brain finds the information, then an affirmative response will be “Yes, it is correct, the team of Martin and John did crush the defending team of in the grand finals of eSports tournament 2024, winning the overall eSports category
 - If no such information exists in the human brain, then a contradicting reply will be “No, it’s wrong, it didn’t happen at all, they got beaten by the defending team”.
- Finding the important context from the statement, in which, it is the question below:

“did they actually crush the defending team during the finals?”

After identifying the most relevant context from the statement, the human mind will answer the question above using the data that was retrieved that is an exact/similar match to the question and the final answer from the statement is told as:

“Yes, it is correct that the team of Martin and John did crush the defending team in the grand finals of eSports winning the overall eSports tournament

A Large Language Model (LLM) is a type of artificial intelligence model designed to understand and generate human-like natural language and it mimics the process of understanding human input in a numerical and computational representation of human input. It is built upon four core components that define how it processes and responds to text-based input. The first is the Transformer Architecture, which serves as the foundational structure of modern LLMs, utilizing encoder-decoder structures to analyze text and allowing for faster training times on GPUs compared to older, sequential models such as RNNs and LSTMs. The second is the Self-Attention Mechanism, which allows the model to weigh the importance of different words in a sentence relative to one another, for example, distinguishing the word "bank" as a riverbank versus a financial institution depending on the surrounding context. The third is Parameters (Weights), which are the internal variables, often numbering in the billions or trillions, learned during training that determine how the model processes input and generates output. The fourth is Tokens, LLMs do not read whole words, but rather subword units called tokens, which allow the model to manage vocabulary efficiently across different languages and writing styles, including code-switched Taglish input.

Among these four components, the Self-Attention Mechanism is the most critical to the operation of Bantanong's generation layer. Its core computational mechanism is formally expressed as:

The Self-Attention formula allows the LLM to dynamically determine which parts of the input are most relevant when generating each word of the response. The process begins with three matrices derived from the input: the Query matrix (Q), which represents what the current token is looking for; the Key matrix (K), which represents what each token in the sequence contains; and the Value matrix (V), which represents the actual content of each token that will be passed forward if deemed relevant.

At each step, the system computes the dot product of Q and K (QK^T), producing a raw relevance score between every pair of tokens in the input sequence. This score is then scaled by the square root of the key dimension ($\sqrt{d_k}$) to prevent the values from growing too large, which would cause the softmax function to produce near-zero gradients and slow down learning. The softmax function then converts these scaled scores into a probability distribution, ensuring all attention weights sum to 1, representing how much focus the model places on each token. Finally, these attention weights are multiplied by the Value matrix (V), producing a context-aware representation of the input that the model uses to generate the next token.

Calculation:

Given a simplified input of three tokens representing a student query:

- $Q = [Q1, Q2, Q3]$
- $K = [K1, K2, K3]$
- $V = [V1, V2, V3]$

Step 1 - Compute raw scores (QK^T):

$$Attention(Q, K, V) = softmax \left(\frac{QK^T}{\sqrt{d_k}} \right) \cdot V$$

- The dot product of Q and K transposed produces a 3x3 matrix of raw relevance scores, where each row corresponds to a query token and each column corresponds to a key token. For brevity, only the computation for the first query token Q1 is expanded below; the same operation repeats for Q2 and Q3 to complete the full matrix:
 - Row 1 (Q1): [Score(Q1,K1), Score(Q1,K2), Score(Q1,K3)] Row 2 (Q2): [Score(Q2,K1), Score(Q2,K2), Score(Q2,K3)] Row 3 (Q3): [Score(Q3,K1), Score(Q3,K2), Score(Q3,K3)]
 - Each score is computed as the dot product of the respective query and key vectors, producing a scalar that represents the raw relevance of that particular key to the given query token. The full 3x3 matrix is then passed to the scaling and softmax steps.

Step 2: Scale by $\sqrt{d_k}$:

- Scaled Score = $QK^T / \sqrt{d_k}$

Step 3: Apply softmax to get attention weights:

- Attention Weights = $softmax(Scaled\ Score) = [w1, w2, w3]$ where $w1 + w2 + w3$
- $= 1$

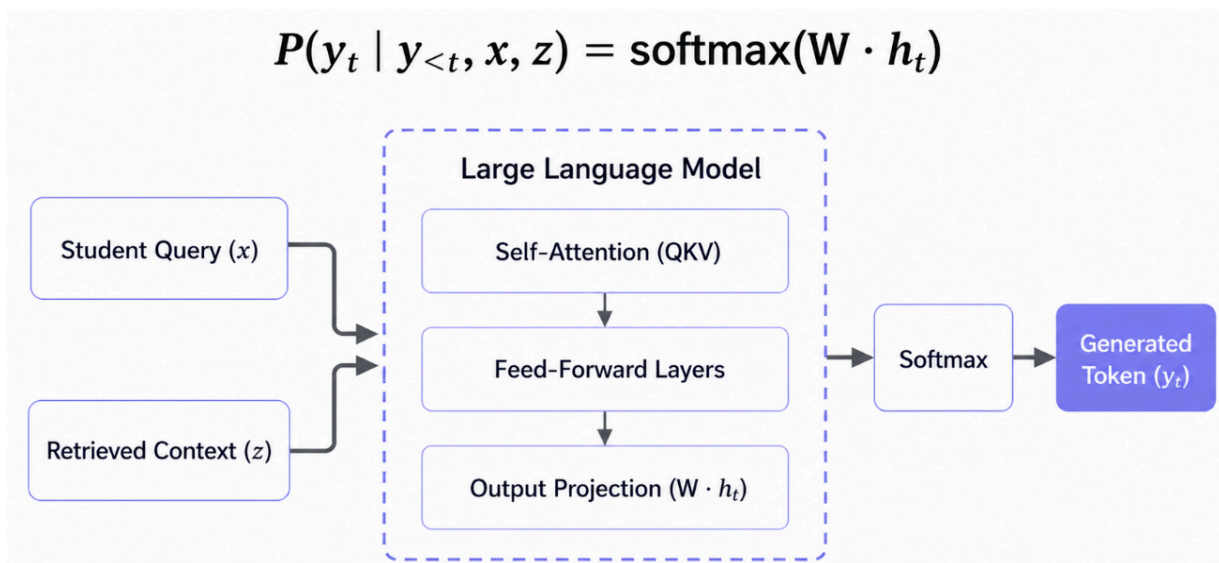
Step 4: Multiply by Value matrix:

$$\mathbf{Output} = w1 \cdot V1 + w2 \cdot V2 + w3 \cdot V3$$

The final output is a weighted sum of the Value vectors, where tokens most relevant to the current generation step receive the highest weights. In the context of Bintanong, this means that when a student asks about enrollment procedures, the self-attention mechanism assigns higher weights to the retrieved policy passages directly describing enrollment steps, and lower weights to unrelated portions of the input, ensuring the generated response is grounded in the most contextually relevant institutional content.

The final token generation is then expressed as:

$$P(y_t \mid y_{<t}, x, z) = \text{softmax}(W \cdot h_t)$$



At each generation step, the final Transformer layer produces a hidden state h_t , which is then projected through the learned output matrix W to produce raw scores over the entire vocabulary. Finally, the softmax function converts these raw scores into a probability distribution, from which the most probable next token is selected as the next word in the response.

Alignment with the Core Research Problem

The Large Language Model serves as the natural language generation layer of Bintanong, responsible for producing fluent and conversational responses that students can naturally interact with. While the KNN algorithm retrieves the most relevant institutional document chunks and the FOL symbolic engine enforces hard-constraint institutional rules, the LLM translates these outputs into clear, human-readable guidance. However, because LLMs are inherently probabilistic, the generated answer does not guarantee correctness, a model may produce a plausible-sounding but factually incorrect response, a phenomenon known as hallucination. This is precisely what necessitates the integration of RAG grounding and FOL symbolic constraints within Bintanong's architecture, preventing the LLM from relying on statistical approximation when answering structured institutional queries. In this proposed study, the LLM component is implemented using a locally deployed Large Language Model, responsible for translating the outputs of the RAG pipeline and FOL symbolic engine into fluent, student-friendly responses.

CHAPTER IV

METHODOLOGY

This chapter outlines the systematic procedures, architectural design, and technical framework to be employed in developing the proposed solution. It translates theoretical concepts into a concrete methodology, detailing the software development life cycle, the integration of the retrieval pipeline, and the operational workflows required to build the system.

Conceptual Framework

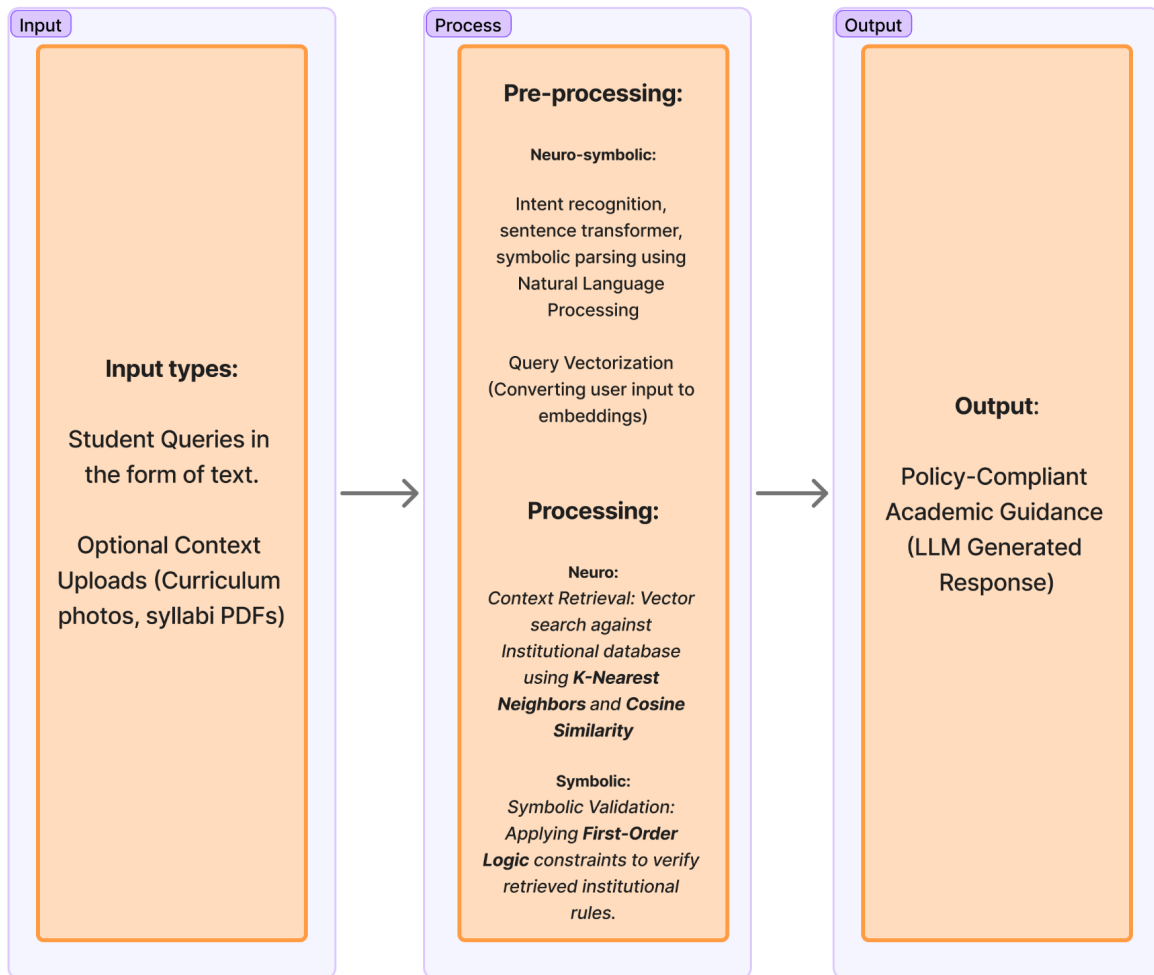


Figure 4.1: The conceptual framework of the study

Figure 4.1 shows the Input-Process-Output framework above, which describes the multi-step pipeline that Bintanong will use to process student queries regarding PalSU policies in their context, from the moment it is sent to the pipeline, designed to transform user queries into a contextualized response, turning their questions into proper AI-assisted guidance for navigating the maze of institutional policies.

Input stage:

During this stage, it covers all data that enters the system, mainly text and document input submitted through a web-based interface, and the institutional knowledge base for system knowledge.

Main input:

The input that the system supports first and foremost are:

Text-based queries:

- The text-based queries that the system expects are mainly written in Filipino and English, or in Tagalog-English language alternation
- These two languages are the common languages spoken by majority of PalSU students, and commonly switch when situation calls for, either in academic, administrative and/or social contexts.

Why Bintanong supports both English and Tagalog languages:

- **It is crucial** for Bintanong's language design consideration, since the existing AI systems currently support the super-majority of languages

such as English and other widely spoken languages, e.g., French, Japanese, and Chinese (Maminta et al., 2025; Bilquise et al., 2022)

- Due to the current super-majority support for the above languages, it is not reflective of how PalSU students speak, which creates a linguistic barrier for better understanding of LLM responses, since the same students are more comfortable in using Tagalog and English and their code-switched combination (Maminta et al., 2025; Bilquise et al., 2022).

How Bintanong would handle:

- **Messy or vague questions**
- **Taglish queries**

To handle the said questions above, Bintanong will:

- Pre-process the questions to its sentence embedding model, capturing everything from the words, down to the syllables and grammar constructs to, which uses SEA-LION-Embedding-E5-600M as the embedding model, so when passing along the message in vector format to the LLM during generation, the entire semantic meaning of the sentence will be preserved.
- For vague questions, it will also enter the same pipeline of , however, once the LLM detects vagueness in its intent, for example:
 - The question **“Can I get an ROG and COG from the same college office”**, while in terms of PalSU institutional linguistics, the question was framed as “can I get both grade reports from the same college office

(usually in deans' office), and ROG was meant as Report of Grades, COG as Copy of Grades, an LLM might trip up since it does not explicitly know what it could mean in institutional context.

For those situations, Bintanong will throw back a series of questions to the user, with a similar implementation to the more recent evolution of artificial intelligence implementations, agent-based AI, (ex: Claude Code, ChatGPT Codex), to gain a better understanding of a users' vague intent, preventing valid but unapplicable answers.

What about Taglish queries?

- Since the thesis will use SEA-LION-Embedding-E5-600M, an embedding model that already was pre-trained and more focused to be used on Southeast Asian languages, its familiarity with the Southeast Asian language semantics helps to properly process the entire query's grammatical properties, preserving the semantics of the query and allowing the NLP during intent recognition to properly understand the user's question to the system.

Secondary input:

The input that the system supports as a secondary complement is:

User-uploaded photos and PDFs -

- These are the documents in either photographic format like .jpg, .png, or in Portable Document Format (.pdf), that may include supplemental context such as, but not limited to:
 - Report/Copy of Grades
 - Prospectuses of undergraduate programs
 - Administrative and academic rules

These files provide additional context for user queries, since Bintanong's institutional knowledge does not fully cover the entire scope of PalSU's institutional rules, including its administrative and academic side.

These files are also processed similarly to the institutional dataset, which is also sent to the same processing pipeline as Bintanong

As a security, privacy, and accuracy measure, those user-provided PDFs and photos will, and should not interfere with the institutional knowledge base of PalSU, as it is fully isolated to the user's chat context, avoiding the possibility of user-provided data entering the institutional database and poisoning it.

The second type of data input for Bintanong is the institutional knowledge base inside PalSU, which consists of the following:

- PalSU Student Handbook
- PalSU Citizens' Charter
- PalSU University and College Student Government (USG/CSG) policies

These are the files that are input and processed before system deployment, where processes that turn these documents into machine-understandable vector values start to appear,

including methods such as vectorization, text chunking, and storage in a vector database for later retrieval.

The documents are the only source of factual information the system initially draws on for each query when asked policy-related questions, and to avoid erroneous outputs, the system does not rely on or use the base language model's out-of-the-box knowledge for this purpose.

Pre-processing stage:

This stage covers the pre-processing stage, where all computational aspects of Bintanong regarding user queries first take place and are pre-processed before reaching the processing stage.

During **pre-processing**:

Assume the user does the following:

- Ask Bintanong about institutional rules (e.g grade policy against their actual course grade)
- Uploads their grade screenshot to their Bintanong chat instance

Sample:

“I failed CC4/L [Data Structures and Algorithms], can I take the next course that requires CC4/L as its prerequisite, let’s say, CC5/L [Information Management 1] or can I not be able to take it?
[uploads their CC4/L grade + prospectus picture]

These queries, alongside optional uploaded documents are:

- Passed through the NLP pipeline to identify the core context, user’s intent.

- The NLP process might determine the following intent as:

CC4/L = FAIL

CC5/L need CC4/L

User want to take CC5/L after FAIL CC4/L ASK IF CAN
TAKE OR NOT

In formalized English:

The user failed CC4/L, and the user stated that it is a needed prerequisite course for many courses, but the user asks if they can take CC5/L, which is a course that requires CC4/L or not be able to take it.

After processing the intent, it is then prepared for further processing, such as text normalization, in case if Bintanong receives a broken text formatting, usually found with text contractions, or misspelled words due to factors like erroneous typing or voice typing, as provided by the example below:

“i faild cc4l can i tak the nxt corse that require cc4l as prereq lets say cc5 or cannot be abl to take it?”

In which, once the intent is determined, it will be treated as the following:

“I failed CC4/L [Data Structures and Algorithms], can I take the next course that requires CC4/L as its prerequisite, let’s say, CC5/L [Information Management 1] or can I not be able to take it?”

Since the sample situation had a user who uploaded a picture, the LLM cannot natively process text input from a picture, so the *optical character recognition (OCR)* component enters

the arena, as the OCR component has the capability to recognize texts as input to the scanned PDFs, and uploaded photos, and in this instance, Docling was chosen as the OCR component, since it performs layout-aware OCR in converting document images, uploaded user images into structured text, with regards to preserving section headers and its tabular structure found within many of PalSU documents such as curriculum documents and PalSU memos

To validate OCR reliability done by Docling itself, the researchers used a small, controlled sample of 10 scanned documents that were manually cross-checked against Docling's extracted text to check for text accuracy, checking for inconsistencies like character errors to unscannable text owing to a bunch of environment factors during data ingest, yielding a character-level accuracy of [X]%, deemed acceptable for downstream chunking and embedding.

After chunking, the text is converted into vector embeddings through a sentence transformer or embedding model. This allows the user query and uploaded context to be represented in the same vector space as the institutional documents stored in the system. Through this preparation, the input becomes ready for semantic similarity search during the neural retrieval stage.

At the same time, relevant rule-based information from the query, such as subjects, prerequisites, academic status, or enrollment-related conditions, may also be extracted for symbolic processing. This prepares the input for later validation using the system's encoded institutional rules. Hence, the pre-processing stage serves as the preparation layer that transforms raw user input into structured, searchable, and logically usable data for the next processing stage.

Processing stage:

In the **processing stage**, the system first classifies the user query as either general, conversational, or institution-specific. General and conversational queries may be handled directly by the LLM, while institution-specific or policy-related queries proceed through the full AI pipeline.

For institution-specific queries, the vectorized query is processed through the neural retrieval layer. This layer compares the query embedding against the institutional knowledge base using vector search. K-Nearest Neighbors is used to identify the top-k most relevant document chunks, while cosine similarity serves as the mathematical basis for measuring semantic closeness between the query and stored institutional documents. The retrieved document chunks are then used as contextual input for the LLM, allowing it to generate a preliminary response grounded in institutional data.

After the preliminary response is generated, the symbolic layer validates the response against encoded institutional rules. This layer checks rule-based elements such as course prerequisites, academic status, enrollment requirements, and policy constraints using First-Order Logic. Through this validation, the system ensures that the LLM-generated response aligns with the institution's official rules and does not rely solely on probabilistic text generation.

Overall, the processing stage combines neural retrieval and symbolic validation. The neural component provides relevant institutional context, while the symbolic component enforces logical and policy-based correctness. This allows Bintanong to generate academic guidance that is both contextually relevant and policy-compliant.

Output:

The output stage produces the final response and shows it to the student, generating the “Policy-Compliant Academic Guidance Response” or the multi-step LLM response to the user context, based on retrieved context and the symbolic logic layer from the activation time.

Since the final response is based on the retrieved institutional data and the symbolic layer, the response is more reliable than the pre-trained knowledge of the base language model, and it allows cross-verifying the response as it also indicates where and which part of the document it was retrieved from for the end-user, i.e., students, to easily cross-verify if needed.

The final responses generated by the system will be evaluated using fundamental statistical measures. The Accuracy, Precision, Recall, and F1-Scores, RAGAS provides a quantifiable measure of the LLM's performance through specific metrics: Answer Relevance checks for off-topiciness; Faithfulness monitors hallucination rates by ensuring outputs match retrieved data; Context Precision verifies that highly relevant items are prioritized in the retrieval sequence; and Context Recall measures if the necessary information was successfully found to construct the final response.

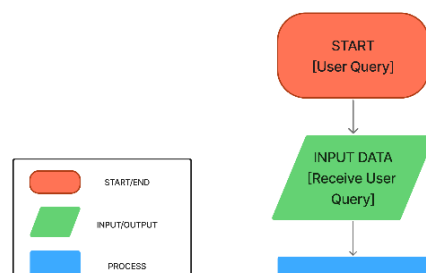


Figure 4.2: System Flowchart

Design and Methodology

The design and methodology of the proposed study provides an outline to the structured approach for developing Bintanong, an AI academic guide that utilizes a neuro-symbolic RAG-based architecture. This section details the flowchart of the system process, data acquisition and parsing of institutional policies, implementation of the concurrent neural and symbolic pipelines, output validation, and performance analysis. The design and methodology will assist the researchers in managing the phases of this proposed study. The findings can contribute to the advancement of hallucination-resistant advisory systems for academic institution

Research Design:

Research Type

This study employs three complementary research types that together address the full scope of developing and evaluating Bintanong.

First, the study is applied research. The primary goal is not to generate abstract theory but to design, build, and deploy a functional AI system that directly solves the documented problem of inaccessible institutional guidance at PalSU. The output of this research is a working web-based platform, not a theoretical model.

Second, the study uses a mixed-methods approach. It combines quantitative performance data, including Accuracy, Precision, Recall, and F1-Scores, RAGAS scores, BATAYAN benchmark accuracy, and SUS numerical ratings, with qualitative data from User Acceptance Testing (UAT) structured test cases. This combination allows the researchers to evaluate

Bintanong from both an engineering performance perspective and a human usability perspective, ensuring the system is both technically accurate and practically usable.

Third, the study is evaluative research. It does not simply describe what was built; it measures Bintanong's performance against pre-defined quantitative benchmarks and real-world usability criteria. The evaluation design is structured to produce measurable evidence that answers each specific research objective

Software Development Model

This study will implement the Agile software development model through iterative sprints. Agile is a flexible, incremental approach to software development in which the system is built in short development cycles called sprints, with each sprint delivering a working and testable component before the next sprint begins. Unlike the Waterfall model, which requires all requirements to be finalized before development starts, Agile allows the development process to respond to changes and new information throughout the project lifecycle.

For Bintanong, each sprint will target one modular component of the system. The first sprint will cover the Taglish NLP pipeline and Intent Analyzer. The second sprint will cover the RAG retrieval pipeline and “**pgvector**” knowledge base construction. The third sprint will cover the SWI-Prolog symbolic reasoning layer. The fourth sprint will cover the integration of all three components through the Adaptive Query Router. Subsequent sprints will cover system-wide testing, evaluation using the defined metrics, and final deployment to the Vercel platform.

Agile is specifically chosen over other software development models for three reasons. First, PalSU's institutional policies are subject to revision during the study period, and Agile's

iterative structure allows the knowledge base and symbolic rules to be updated between sprints without disrupting the entire pipeline. A Waterfall approach would require the knowledge base to be finalized before development begins, which is impractical given that policy documents are still being sourced from their respective offices. Second, the modular architecture of Bintanong, which separates the NLP layer, the RAG retrieval layer, and the symbolic reasoning layer, naturally fits a sprint-based delivery model where each module is independently built, tested, and refined before integration. Third, Agile accommodates continuous student feedback throughout the development process, which is important for a system whose usability is one of its primary evaluation criteria. Feedback gathered during early sprint reviews can be incorporated into later sprints before the final system is evaluated.

Ethical Considerations

The development and evaluation of Bintanong involve collecting data from human participants and deploying an AI system within an institutional academic environment. The following ethical protocols will govern all aspects of the research to ensure the protection of participants, the integrity of the study, and the responsible use of artificial intelligence.

Informed Consent

All student participants in the usability and user acceptability evaluation will receive an informed consent form before the evaluation session begins. The form will explain what the study is about, what participants will be asked to do, what data will be collected, and how it will be used. No student will be included in the evaluation without first signing the consent form. Participants will also be informed that they can withdraw from participation at any time without any effect on their academic standing.

Data Privacy

The development and operation of Bintanong will fully comply with the Data Privacy Act of 2012 (Republic Act No. 10173), which governs the collection, processing, and protection of personal information in the Philippines. Under RA 10173, personal information must be collected for specified and legitimate purposes, processed in a manner consistent with those purposes, and protected against unauthorized access, disclosure, or misuse. In line with these provisions, the researchers will ensure that no personal data beyond what is strictly necessary for system evaluation will be collected from participants.

Each student's conversation with Bintanong will be stored and processed independently within an isolated user chat context. No student will be able to access or influence another student's chat session. User-provided photos and PDFs uploaded during a session will be fully isolated within that user's context and will not be added to the institutional knowledge base. Lastly, no personal academic records, including grades, enrollment status, or student identification, will be stored by the system. These measures are consistent with the data

minimization principle under RA 10173, which requires that only the minimum necessary personal information be processed in relation to the stated purpose of the research.

Read-Only System Architecture as an Ethical Safeguard

Bintanong is designed to be a read-only advisory tool. It does not have access to any university database and cannot modify student records, enroll students, or perform any administrative transaction on a student's behalf. This architectural boundary is not only a technical design decision; it is an ethical safeguard that ensures the system cannot inadvertently alter any official student information regardless of how it is used.

Responsible Use of Artificial Intelligence

Since Bintanong is an AI system deployed in an academic guidance context, the research team recognizes its responsibility to minimize the risk of providing students with incorrect institutional information. The Neuro-Symbolic RAG architecture is designed with this responsibility in mind: the RAG pipeline ensures responses are grounded in the official documents, and the First-Order Logic symbolic layer enforces hard institutional rules such as prerequisite dependencies. Even with these safeguards, the system interface will display a visible disclaimer informing students that Bintanong is an AI guide and that important academic decisions should still be verified with the relevant university office or academic advisor.

Institutional Document Authorization

The four documents that will constitute Bintanong's knowledge base, namely the PalSU Student Handbook, the PalSU Citizens' Charter, and the USG and CSG Policies, are publicly available official documents of Palawan State University and its recognized student governance

bodies. Because these documents are designated for public access in accordance with government transparency guidelines, such as the Anti-Red Tape Act (ARTA), explicit written authorization is not legally required for their inclusion in this academic research. However, to ensure the highest degree of data accuracy and as a professional courtesy, the research team will submit a formal letter of request to each issuing authority. This includes the Office of Student Affairs and Services for the Student Handbook, the University Planning Office for the Citizens' Charter, and the USG and CSG offices for their respective policies. These letters will inform the offices of the system's development and request the official digital copies of their documents. If any document is updated during the study period, the latest version will be requested through the same communicative process to ensure the knowledge base remains current.

Participants and Dataset

For the research's participants, it will be limited to the following set of participants:

1. Undergraduate students of Palawan State University at the following colleges:
 - a. College of Architecture and Design
 - b. College of Arts and Humanities
 - c. College of Business and Accountancy
 - d. College of Criminal Justice Education
 - e. College of Engineering
 - f. College of Hospitality and Tourism Management
 - g. College of Nursing and Health Sciences
 - h. College of Sciences
 - i. College of Teacher Education

For the research's dataset:

The second type of data input is the institutional knowledge base inside PalSU, which consists of the following:

- PalSU Student Handbook
- PalSU Citizens' Charter
- PalSU University and College Student Government (USG/CSG) policies

These are the files that are input, processed before system deployment, where processes that turn these documents into machine-understandable vector values start to appear, including methods such as vectorization, text chunking, and storage in a vector database for later retrieval.

These documents are the only source of factual information the system initially draws on for each query when asked policy-related questions, and to avoid erroneous outputs, the system does not rely on or use the base language model's out-of-the-box knowledge for this purpose

Each query will be written in Taglish to reflect how PalSU students naturally phrase institutional questions, and each will be paired with a verified correct answer taken directly from the official PalSU documents.

This dataset will serve as the ground-truth reference for computing Accuracy, Precision, Recall, and F1-Scores, RAGAS scores and comparing the full Bintanong pipeline against the baseline Bintu-1 model.

System Evaluation Dataset

The primary dataset for system performance evaluation will be a benchmark query dataset constructed by the researchers.

This dataset will consist of at least fifty (50) institutional queries distributed across five categories: enrollment procedures, prerequisite eligibility, grade and academic standing, and disciplinary policies.

Why 50 institutional queries was chosen for Bintanong:

In the categories that Bintanong covers for an institution like Palawan State University such as enrollment procedures, prerequisite eligibility, grade policies, academic standings and disciplinary policies, the researchers find it exhaustive to cover all bases of such categories, so the researchers will consolidate the 10 most common queries in each of the categories, as not only it is the commonly asked questions inside the university by students itself, but the common questions have a common answer to be readily used as a 1st line of reference, which is substantial for creating a validation sample like Bintanong would be aiming for, when designing a Neuro-symbolic RAG system for academic context.

As noted by Lugones et al. (2026), the study, Aurora, was already benchmarked using a 20-query benchmark that covers similar bases as Bintanong, such as next-semester course selection in comparison to grade history, and its out of scope questions, in order to check for the system's complete reasoning chain and to provide a chain of evidence for its factual correctness and quality level of explanation.

As noted by the same study, each benchmark query inside Aurora was also followed with a verified, expert reference answer to verify its factual and logical truth to the institutional constraints that was applied to Aurora.

In this case, Aurora's approach to proving its logical and factual correctness is IN LINE with Bintanong's overall approach in testing its entire system pipeline against the chosen series of common questions. Lastly, it also makes it easier to test for Bintanong as the same study already uses that same framework in approaching its logical and factual soundness for its system generated answers.

Knowledge Base Documents

The institutional knowledge base that Bintanong will draw from consists of three official PalSU documents. The first is the PalSU Student Handbook, which contains the university's academic policies, disciplinary rules, prerequisite structures, graduation requirements, and student rights. The second is the PalSU Citizens' Charter, which provides step-by-step procedures for administrative transactions that students can request, such as obtaining a Certificate of Grades or processing an excuse letter. The third is the USG and CSG Student Policies, which govern the student government rules and procedures applicable to undergraduate students.

User Evaluation Participants

The participants for the usability and user acceptance evaluation will be undergraduate students from PalSU Main Campus. Participants will be selected through purposive sampling to ensure that all evaluators represent the exact target users of the system. While the System Usability Scale (SUS) does not mandate a specific sample size, as it is a questionnaire instrument rather than a study protocol, thirty (30) participants is adopted as the minimum threshold to ensure score reliability while remaining feasible within the study's timeline and participant availability at PalSU Main Campus (Hertzum, 2026). Participants for the User Acceptance

Testing (UAT) will be undergraduate students who will interact with Bintanong under conditions that closely simulate actual use.

Research Instruments

The study will use five data collection instruments organized into two groups: instruments for evaluating system performance, and instruments for evaluating user experience.

System Performance Instruments

To analyze and evaluate Bintanong's performance is imperative to validate its functional efficacy and the accuracy of its information retrieval when responding to student inquiries. Researchers will construct a benchmark query dataset of at least fifty (50) institutional queries distributed across five categories: enrollment procedures, prerequisite eligibility, grade policies , academic standing, and disciplinary policies . Each query will be paired with a verified correct answer taken directly from the official documents. To ensure a comprehensive, rigorous, and contextually appropriate assessment, this research will utilize a multi-tiered statistical method comprising the following metrics :

(1) Accuracy, Precision, Recall, and F1-Scores to measure the fundamental objective functionality and retrieval accuracy of the system. This study will employ Precision, Recall, and F1-scores. These statistical metrics are standard measures for evaluating the operational accuracy of classification and retrieval models, ensuring that the system reliably extracts the correct institutional policies. The application of these metrics aligns with established methodological frameworks in recent PalSU AI advisory and pipeline studies (Abela et al., 2025; Cosicol et al., 2023; Lugones et al., 2026).

(2) Retrieval-Augmented Generation Assessment (RAGAs) to assess the quality of the system's generated text and mitigate the risk of providing incorrect policy information, the RAGAs framework will be utilized. As detailed by Es et al. (2024), RAGAs is specifically designed to conduct hallucination checks, measure answer relevance, and determine context precision. RAGAs is closely aligned with human judgment, its integration ensures that Bintanong's outputs remain strictly faithful to the retrieved institutional documents and highly relevant to students' specific administrative queries.

(3) BATAYAN Benchmark, given the localized linguistic environment of PalSU, evaluating Bintanong's vernacular and bilingual competence is critical. The system's natural language processing capabilities will be assessed using BATAYAN, introduced by Montalan et al. (2025) as a holistic benchmark specifically tailored for Filipino and Taglish NLP. This framework measures the model's understanding, causal reasoning, and text generation within a local context. Employing BATAYAN directly evaluates Bintanong's capability to accurately process, interpret, and respond to authentic student queries phrased in the vernacular or naturally code-switched language.

User Experience Instruments

The participants for the usability and acceptance evaluation will be undergraduate students from the PalSU-Main Campus using two evaluation standards.

(1) For user usability, researchers will select through purposive sampling. While System Usability Scale (SUS) does not mandate a specific sample size as it is a questionnaire instrument rather than a study protocol. Thirty participants is therefore adopted as the minimum threshold to

ensure score reliability while remaining feasible within the proposed study's timeline and participant availability at PalSU-Main Campus. SUS is a widely used instrument for measuring perceived usability that complements performance measures (Hertzum et al., 2026).

(2) For user acceptance, the proposed study will implement User Acceptance Testing (UAT) to verify that the system satisfies the specific academic and administrative requirements of the end-users. UAT comes from the software development lifecycle (SDLC) and is intended to test how the system would perform in circumstances similar to those in which the system will eventually be used (Gordon et al., 2022).

Procedures

To obtain the necessary data for building the knowledge base required for Bintanong, the data sources, once properly identified, will be requested from the appropriate offices and university/college student government organizations within PalSU. With a proper data acquisition chain, the researchers will adhere to the university's data acquisition policies, thereby preserving the ethical integrity of the data collection process.

To recap, the following data sources will be acquired from each respective PalSU office and organizations:

- The Office of Student Affairs and Services, concerned with student affairs as a “whole university”, such as student policies, will be requested by the researchers for a copy of the **Student Handbook**, a definitive guide on student-related policies regarding academic and academic-adjacent affairs (i.e, grade policies, dress code, student events, and policy violations)

- The University Student Government or USG, concerned with the student affairs that govern the locality of the entire university student body such as student government affairs (i.e, student body constitution), will also be requested to provide a copy of their policies and documents that affect the university's student body, such as, but not limited to:
 - USG's Constitution and By-Laws, accounting for their base structural format for the College Student Government's constitution and by-laws.
 - The College Student Government or CSG, concerned with the student affairs that govern the locality of their colleges, such as student-related affairs (i.e, student activities, and membership), will also be requested to provide a copy of their policies and documents that affect the colleges' student body, such as, but not limited to:
 - CSG's Constitution and By-Laws, accounting for their specific collegiate operational processes.
 - CSG's Student Forms (including but not limited to clearance forms)
 - Other CSG-related student policies and operational rules that CSG members may identify with themselves.

The Citizens' Charter, which outlines the step-by-step procedures for all administrative offices inside the university, will be obtained through multiple steps or options:

- The researchers will obtain a digital Citizens' Charter by accessing the university's digital copy of the Citizens' Charter through its website (<https://psu.palawan.edu.ph/citizen-charter/>), and downloading the PDF copy of the

Citizens' Charter.

- If the digital copy lacks specific, up-to-date student-facing charter data for some administrative offices and/or external units, such as colleges, the researchers will send a request letter to the University Planning Office to request an updated digital Citizens' Charter copy.
- To complement the university's Digital Citizens' Charter, the researchers will visit the student-specific administrative and external units (e.g., the University Library, NSTP Office, and colleges) and photograph the ARTA-required Citizens' Charter posters posted outside their offices.

The researchers have outlined multiple steps and options in obtaining the augmented data for the Digital Citizens' Charter in order to take into account that the Digital Citizens' Charter may or may not completely document all of the student-facing processes

All three documents are currently being sourced from their respective offices. The research team will request formal written approval from each issuing authority, signed by the adviser and the chairperson, before any document is digitized or indexed into the system.

This includes the Office of Student Affairs and Services for the Student Handbook, the relevant administrative office for the Citizens' Charter, the USG office, and the CSG office for the CSG Policies.

Once approval is obtained and communicated to the researchers, the obtained documents will be converted into machine-readable text and prepared for chunking and embedding to build the knowledge base.

Implementation

To fully realize the architectural pipeline for Bintanong, the researchers have opted for the following technologies after careful consideration of performance, ease of implementation, and future maintainability.

The researchers have opted to use “**free and open source software**” as much as possible in the development of Bintanong to demonstrate its maintainability, ease of implementation, and performance. Moreover, the researchers have opted to use “**industry-standard**” tools like Python and PostgreSQL as an example, in order to constrain the development in real-world scenarios.

The following **software and language stack** was chosen by the researchers to develop the AI pipeline for Bintanong:

Software/language/model stack	Software/language/model version	Why was it chosen for Bintanong
Python	3.13.4	Python is one of the most supported languages for AI development, with many AI libraries supporting Python-majority workflows
Next.js	16.2.6	Next.js is a frontend framework that serves as the front-end interface layer and integrates cleanly with Vercel’s AI SDK.

Supabase (PostgreSQL with pgvector)	PostgreSQL 17	Supabase provides a unified platform offering both relational and vector database capabilities. This allows Bintanong to power the RAG mechanism via pgvector while maintaining a structured relational schema for pre-requisite chains and student-related data in a single environment.
SWI-Prolog (via Janus, a Python to Prolog framework)	version 10.1.6	Prolog is a logical programming language that acts as the symbolic layer representation for Bintanong.
FastAPI	v0.136.1	FastAPI is the API handler for the AI pipeline and exposes the pipeline to the frontend for Bintanong.
Vercel AI SDK	6.0.177	Vercel AI SDK handles frontend state for Bintanong and processes state management from the FastAPI pipeline.
“Bintu-1” Large Language Model	Google Gemma4 e4b	Bintu-1 is the designated local model name for Google Gemma4 e4b to fit the overall Bintanong nomenclature. The model was chosen by the researchers for its ability to do good “chain-of-thought” and inquisitive reasoning in its base form.
SEA-LION-Embedding-E5-600M	2026	The researchers chose this sentence embedding model since it converts the Taglish inputs into the same shared vector space without sacrificing the semantics for each input converted to a vector.
Docling	2.93.0	The researchers chose this local OCR platform since it handles the required conversion of text from photos uploaded by the users and for indexing the photos of academic procedures
shadcn/ui	4.2.0	shadcn/ui is the open-source UI (user interface) component library, chosen as a

		free, open UI component for Bintanong.
Tailwind CSS	4.2.4	Components of shadcn/ui use Tailwind CSS, as its implementation is class-based; instead of creating separate CSS files, the code calls Tailwind CSS classes directly.

Table 4.1: Software and Language Stack

Bintanong will be developed on local hardware first to test its constraints and simulate university-grade laboratory hardware in a typical local AI pipeline development scenario, and then deployed in an online virtual environment once it is ready for general use.

The following **hardware stack** was chosen by the researchers to locally test and develop the AI pipeline for Bintanong:

Hardware information	Hardware specification	Why was it chosen for Bintanong
Intel Core i7-12650H (CPU)	2.70 GHz base clock, up to 4.70 GHz boost clocks, 10 cores (6 P-cores, 4 E-cores)	The Intel CPU was chosen to provide ample processing capacity for Bintu-1 in cases where GPU inference fails.
32GB of DDR5 (RAM)	DDR5, SODIMM, 4800 MT/s	32GB of memory was chosen as the safe zone for Bintanong's development computer to account for other development tools running parallel, such as PostgreSQL DBs and virtual environment instances.
NVIDIA GeForce RTX 4060 Laptop GPU (Graphics Card)	8GB GDDR6 of VRAM (Video RAM)	This GPU represents the average card used for local AI inference, where Bintu-1 comfortably fits without maxing out VRAM and using system RAM as fall back.

Table 4.2 Hardware Stack

For the future deployment of Bintanong to an online platform, Vercel will be used to deploy Bintanong to an online server, since the researchers have found that Vercel offers a generous free tier for its online platform to host the research itself, simulating a small, grassroots AI platform in PalSU that addresses the research problems in a smart, thrifty approach. The databases hosted in PostgreSQL format (including pgvector), Supabase will act as the online counterpart for PostgreSQL.

Work Plan

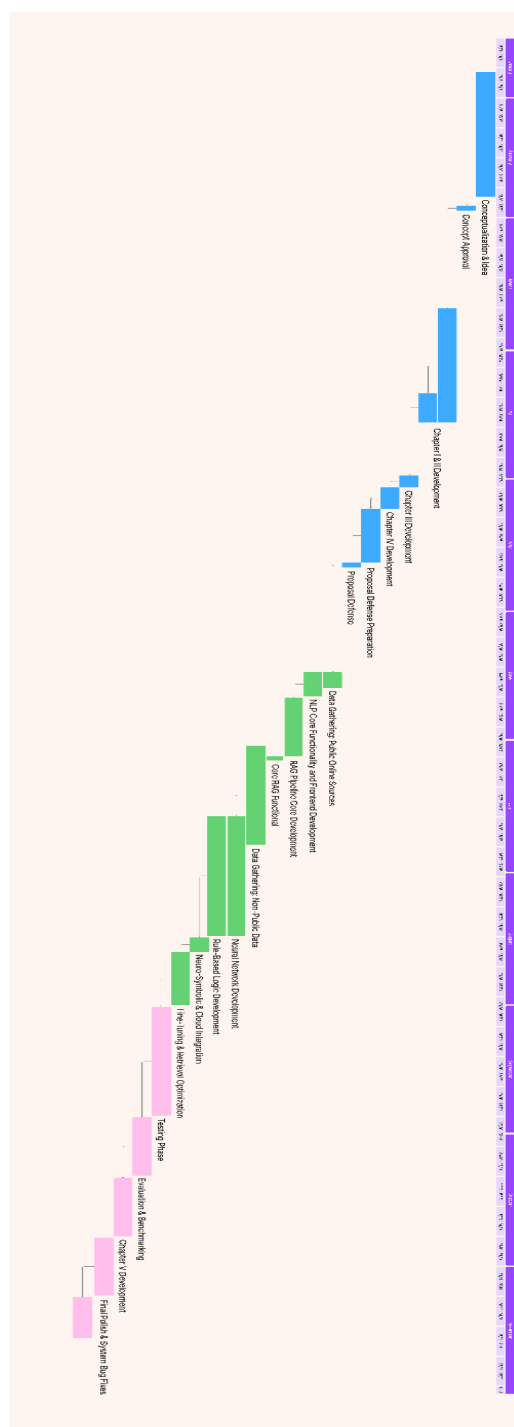


Figure 4.1: Work Plan

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